

2024



South East Water Identifies and Mitigates Water Pressure Anomalies (Transients) using AI

Learn how this Melbourne metropolitan water corporation, established by the Victorian Government in Australia is improving infrastructure monitoring and water management using the TD•Cloud solution powered by AWS.

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Near real-time

monitoring capabilities achieved

>97.5%



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99%

Positive correlation of transient stress



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Root-cause identification



Overview

As demands on its infrastructure continued to grow, [South East Water](#), a Melbourne metropolitan water corporation, established by the Victorian Government in Australia, sought an innovative approach to monitoring its pipe network. In collaboration with data analytics agency [SpiralData](#), an [AWS Partner](#), they trialed the Artificial Intelligence (AI)-powered [TD•Cloud](#). Built on [Amazon Web Services \(AWS\)](#), the platform monitors and analyses fast-moving pressure fluctuations that cause sustained pipe fatigue across focus areas of the utility's 27,000km pipeline network.

This strategic initiative not only enhanced the water corporation's operational efficiency but also strengthened the resilience of water services for over 1.8 million residents in Melbourne's south-east. By trialing this technology, South East Water was able to calm the water network, thereby reducing the potential of service disruptions while extending the lifespan of its pipeline network. Furthermore, this initiative also supports environmental sustainability by minimizing potential non-revenue water loss.



Opportunity | Advancing Pipe Durability with AI-Driven Analytics

Drinking water pipe infrastructure for city development is often designed to last about 100 years. But as it ages, the associated challenges increase. South East Water has been maintaining some of its underground infrastructure for over 50 years and was seeking innovative ways to maintain, prolong, and de-risk the infrastructure.

High pressure water transients can be difficult to detect and diagnose. These are abnormal fluctuations in water pressure that stress the pipe network, causing microtears that reduce pipe durability. Over time, these tears can lead to full pipe bursts, disrupting water services and necessitating earlier capital investments in pipe replacements. South East Water wanted a solution that allowed the utility to proactively monitor and identify these high pressure water transients.



In response, in 2023, South East Water engaged SpiralData through the [Intelligent Water Network \(IWN\)](#) for trialing their TD•Cloud technology. IWN is a collaborative network of water industry professionals dedicated to seeking cutting-edge innovations in water management.

In August 2023, the utility and SpiralData commenced trials of the AI-powered TD•Cloud. This innovative system monitors transients in near real time from high-speed pressure sensors and uses AWS-hosted machine learning (ML) to deepen its analysis and improve the accuracy of its predictive maintenance strategies.

Solution | Optimizing Operations with AI-powered Solutions

South East Water is transforming its approach to water network management by trialing the TD•Cloud solution. Historically, South East Water’s ability to detect fast-moving pressure transients has been limited, making it difficult to pinpoint the root causes of leaks and breaks. With TD•Cloud, the water utility can monitor pressure transients with >97.5 percent detection accuracy. This means strong positive correlations can be established between pressure transient activity to leaks and breaks. This helps in prioritizing at-risk pipes with an accumulative stress score, and isolates the root causes of pipe fatigue across water assets and their customers.

Built on AWS, the TD•Cloud solution features a real-time dashboard for monitoring and analyzing pressure transient patterns. This streamlines decision-making across various management and operational teams by serving as a single source of truth.

SpiralData used [Amazon SageMaker](#) to build, train, and deploy Machine Learning algorithms that process sensor data with high precision. These algorithms have achieved more than 97.5 percent accuracy in detecting pressure anomalies. With these data-driven insights, South East Water can confidently enhance pipeline integrity and optimize maintenance schedules, substantially improving infrastructure resilience. During the technology trials, TD•Cloud successfully identified critical pump assets and industrial customers generating significant pressure transients at-risk pipes, bolstering preventive maintenance efforts to calm the water network.



[Amazon Simple Storage Service \(Amazon S3\)](#) and [Amazon Athena](#) are also integral to the solution's functionality. Amazon S3 is an object storage service that stores vast amounts of data collected. Amazon Athena is a serverless interactive analytics service built on open-source frameworks that processes this data in real time.

As Amazon S3 and Amazon Athena are fully managed services, SpiralData can also focus its resources on enhancing the AI algorithms on the solution. This focus helps SpiralData's customers make rapid, well-informed decisions.

Outcome | Scaling Insights for Future Water Security

Since March 2024, South East Water has been actively working with SpiralData to explore the scalability of the TD•Cloud platform across a broader segment of the utility's pipeline network focusing on 'hot' spots that have a history of leaks and breaks. Through limited trials, sensor deployments and the associated data sets, a strong correlation has been uncovered between the accumulative stress of pressure transients and pipe leaks and breaks. This has increased the value of the platform to refine maintenance strategies and preemptive measures.

Looking ahead, South East Water is assessing the viability and feasibility of an enterprise-wide rollout of the platform. This expansion aligns with the utility's broader vision to transform water and wastewater management across its service area, using data-driven insights to ensure long-term financial sustainability and strengthen water resilience.

"With the TD•Cloud solution, we've taken a major leap forward in **trialing technical solutions** for water network management. This initiative not only showcases the incredible possibilities of combining AI with high-speed sensors but also allows us to actively enhance the safety and reliability of our water supply. I am profoundly grateful for the support of SpiralData, IWN and AWS, which has been crucial in our mission to deliver efficient and sustainable water services to our community. Together, we are setting new standards for the industry," said Ninad Dharmadhikari, Product Development Manager (Systems), South East Water.

Learn More

To learn more, visit aws.amazon.com/energy-utilities.

About South East Water

South East Water is a metropolitan water corporation operating in Melbourne's south-east, established by the Victorian Government in Australia. They support healthy and liveable communities by delivering water, sewerage and recycled water services to almost 1.8 million people (around 30% of Melbourne's population) every day and every night.

AWS Services Used



Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon Athena

Amazon Athena is a serverless, interactive analytics service built on open-source frameworks, supporting open-table and file formats. Athena provides a simplified, flexible way to analyze petabytes of data where it lives.

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